# Analytics & Big Data

Session 9: Big data 2

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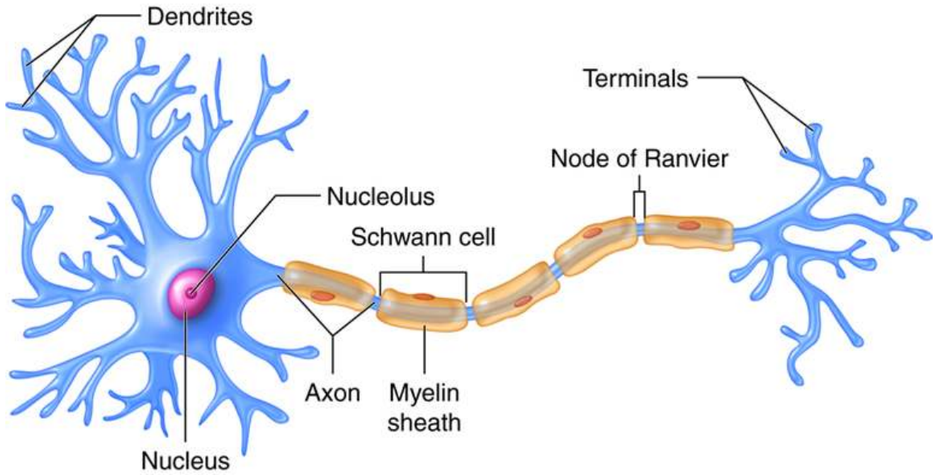
### 🎯 Learning objectives

- **Explain and interpret** the core ideas and mathematical operations underlying artificial neural networks
- **Distinguish** between major neural network architectures (MLP, CNN, RNN, TM) and identify the types of data for which they are most appropriate.
- **Understand** the design and training process of neural networks, including architecture selection and optimization.

# Foundations

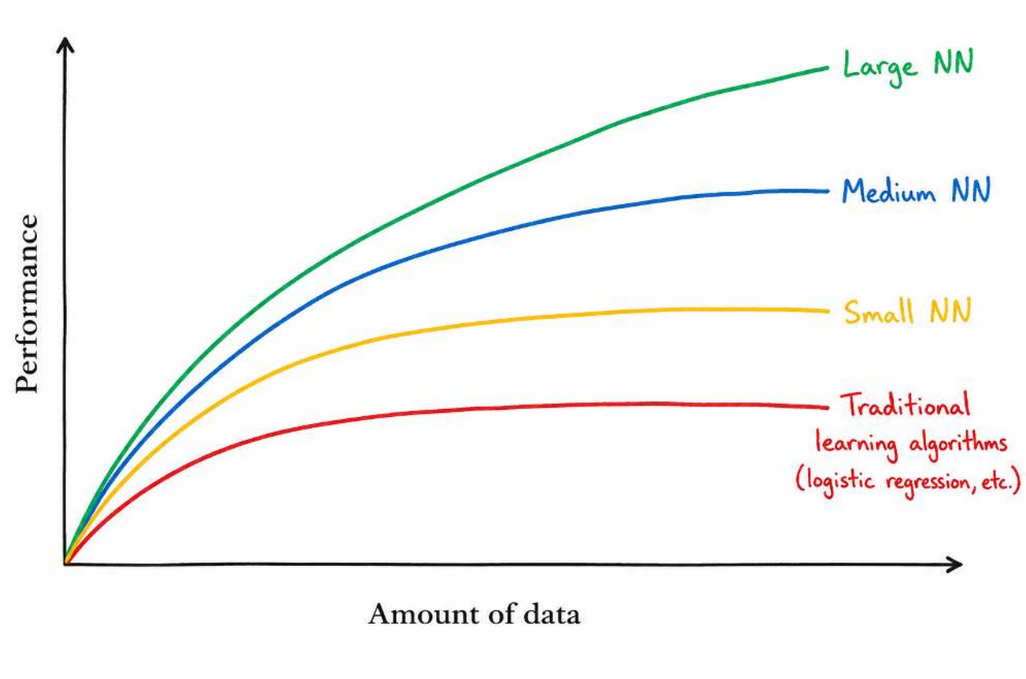
# From biological neurons to artificial neural networks




**Key milestones**

- **1943** (Warren McCulloch & Walter Pitts) — Mathematical model of artificial neurons
- **1958** (Frank Rosenblatt) — Development of the *Perceptron*
- **2010s** (Geoffrey Hinton, Yann LeCun & Yoshua Bengio) — Revival of modern deep learning
- **2020s** (GPT models, OpenAI) — Emergence and widespread adoption of large language models (LLMs) and multimodal AI systems

# Performance of machine learning and neural networks




**Key drivers of recent progress in neural networks**

- Availability of large-scale datasets
- Increasing computational power (especially GPUs/TPUs)
- Advances in algorithms and neural network architectures
- Specialization of neural networks:
  - Convolutional Neural Networks (CNNs) for image and video recognition
  - Recurrent Neural Networks (RNNs) for sequential and time-series data
  - Transformers for language processing and generative AI
  - Graph Neural Networks (GNNs) for network and relational data
  - Diffusion Models for image and media generation

In small-data regimes, experiments are needed to identify algorithms that perform best for a given dataset.

# The perceptron

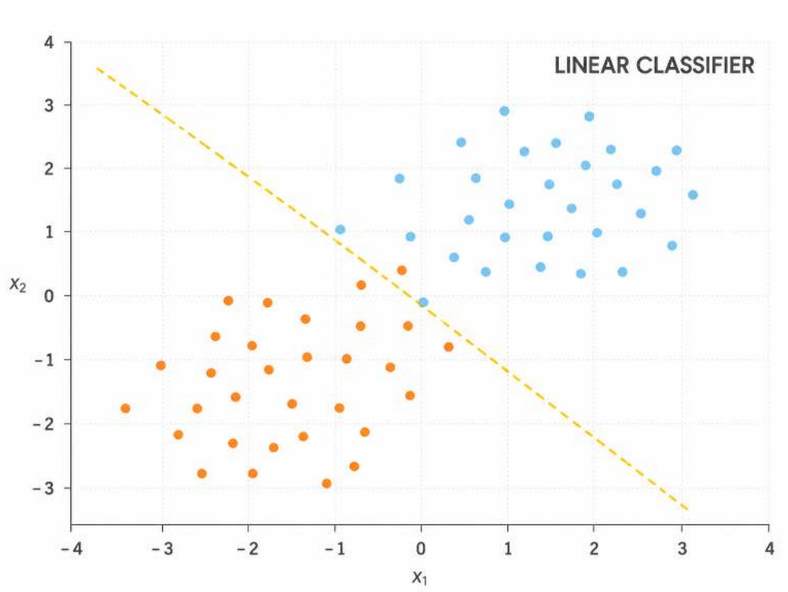
A perceptron is one of the earliest forms of neural networks. A **classical perceptron** is characterized by:

$$z = w^T x + b$$

followed by a **threshold activation**:

$$\hat{y} = \begin{cases} 1 & \text{if } w^T x + b > 0 \\ 0 & \text{otherwise} \end{cases}$$

- This corresponds to a linear classifier
- The training of the perceptron consists of feeding it multiple training samples and calculating the output for each of them
- After each sample, the weights $w$ are adjusted to minimize a loss function, which quantifies the difference between the predicted and true outputs



# From perceptron to deep neural networks

Limitation of a single perceptron: It cannot build abstract intermediate representations and can only learn linear decision boundaries.

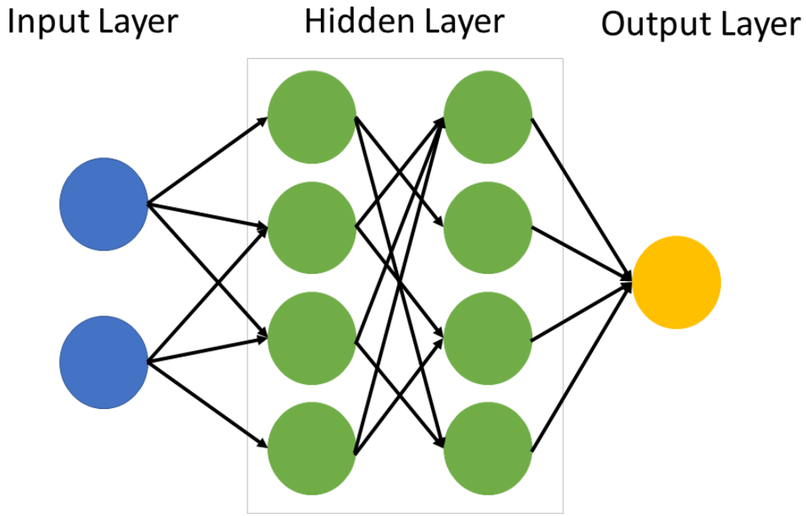
Solution: Multilayer neural networks combine:

- **Multiple hidden layers**, which enable increasingly abstract representations
- **Activation functions**, which make nonlinear learning possible

Layers:

- Input neurons: raw data (e.g., images, audio, video)
- Hidden neurons: the weights and biases store abstract representations learned from the data (e.g., edges, parts of an image)
- Output neuron: prediction (e.g., one neuron for binary, multiple for multi-class)

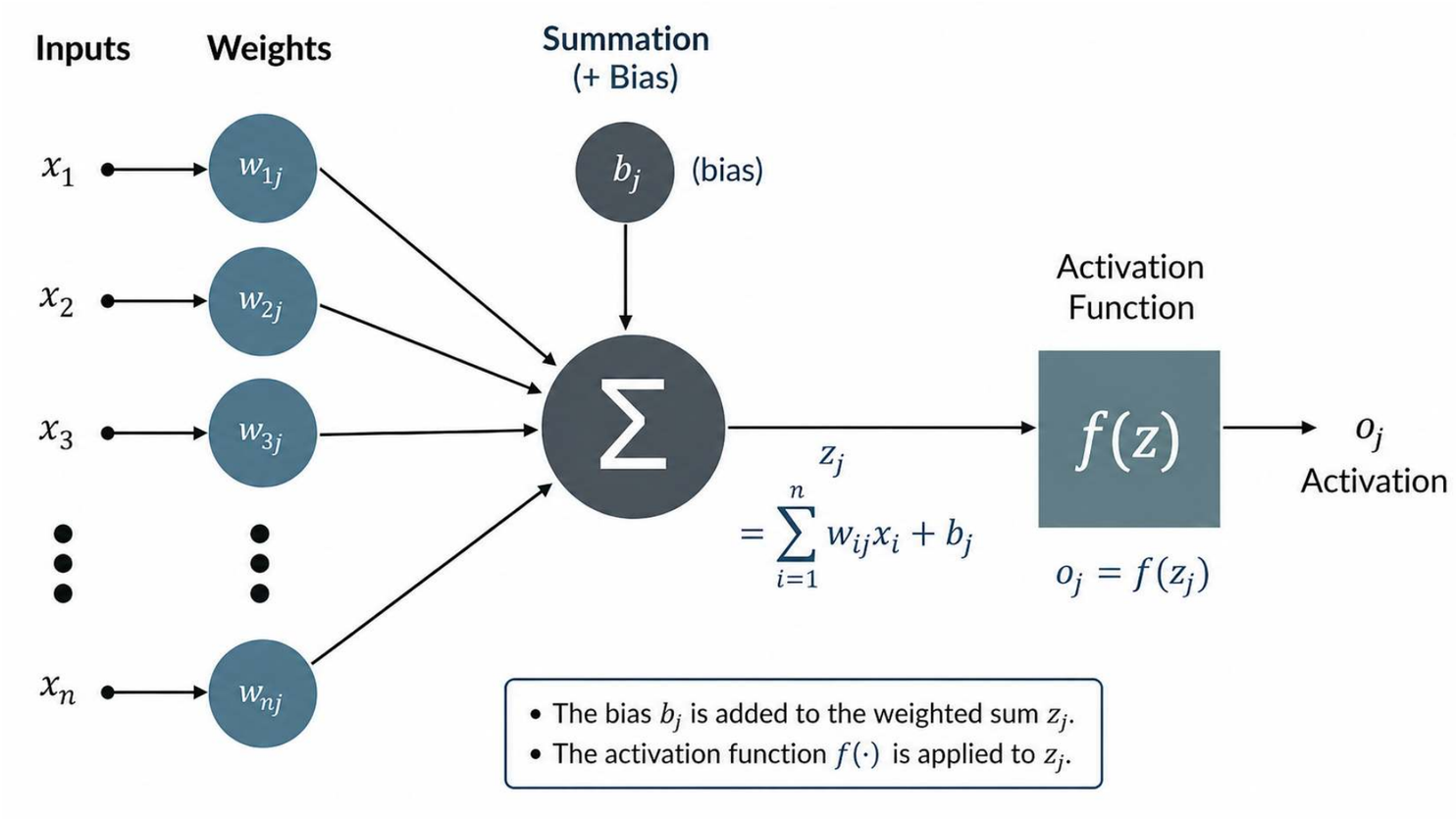
Note: the number of layers and neurons is specified ex-ante and does not change in the training process.




Artificial Neural Networks

🎥 Grant Anderson (3Blue1Brown) provides a series of instructive animated explanations, including one on Neural Networks.

# Functionality of a neuron

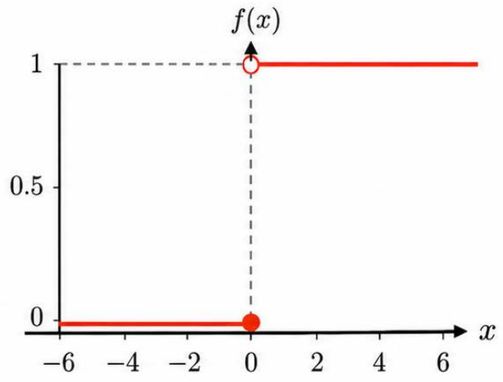



In a classical perceptron, the activation is typically binary (0 or 1), representing whether the neuron “fires.”
In modern neural networks, activations are usually continuous-valued scalars determined by the activation function.

# Types of activation functions

**Step**

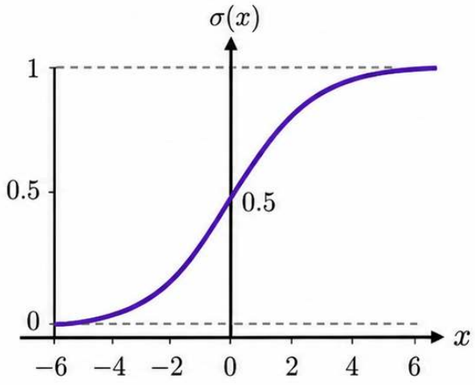
$$f(x) = \begin{cases} 0, & x \leq 0 \\ 1, & x > 0 \end{cases}$$




Note: Perceptron

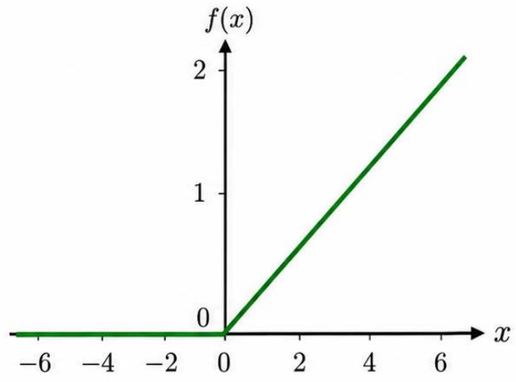
**Sigmoid**

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$




**ReLU**

$$f(x) = \max(0, x) = \begin{cases} 0, & x \leq 0 \\ x, & x > 0 \end{cases}$$

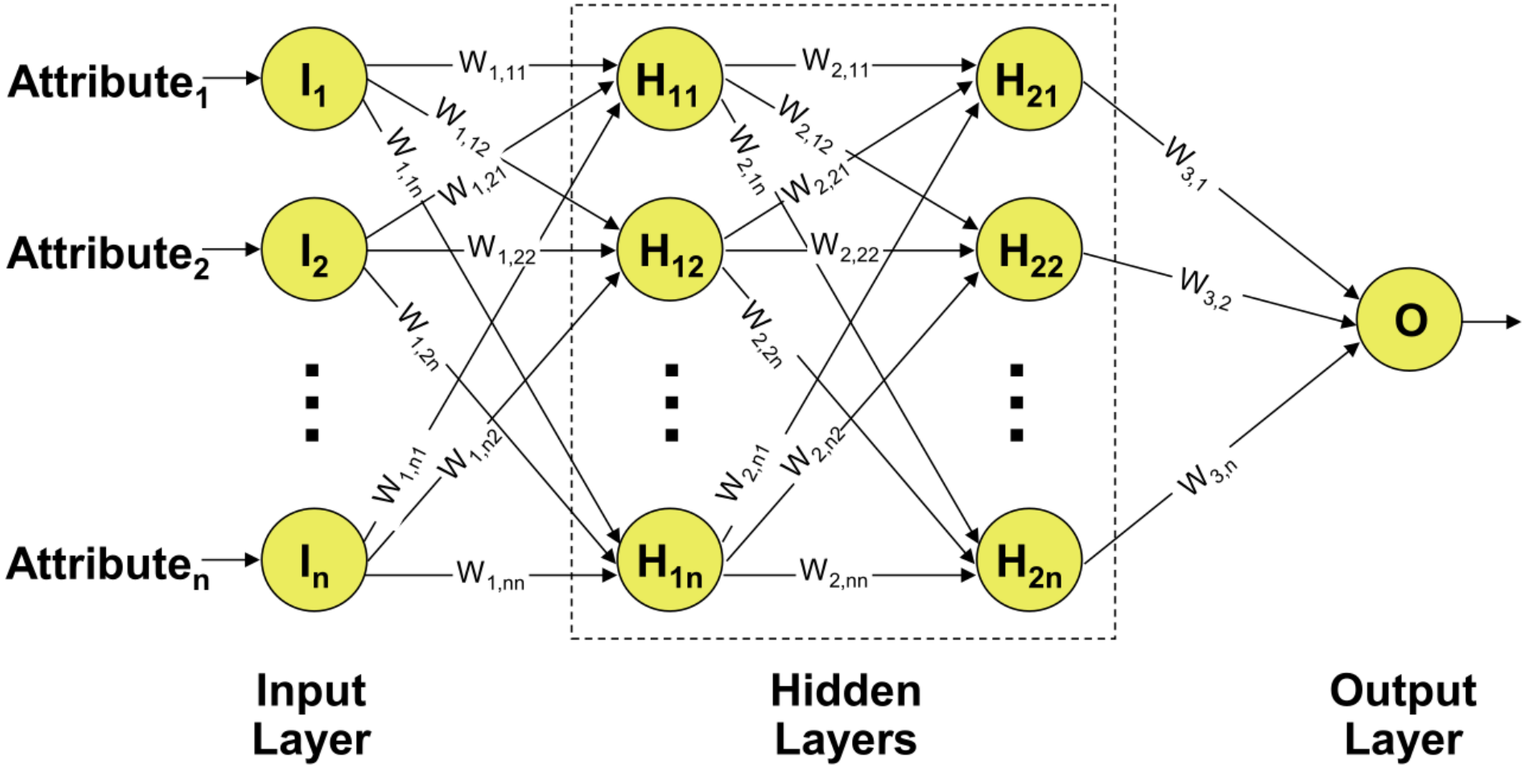



Note: Efficient computation

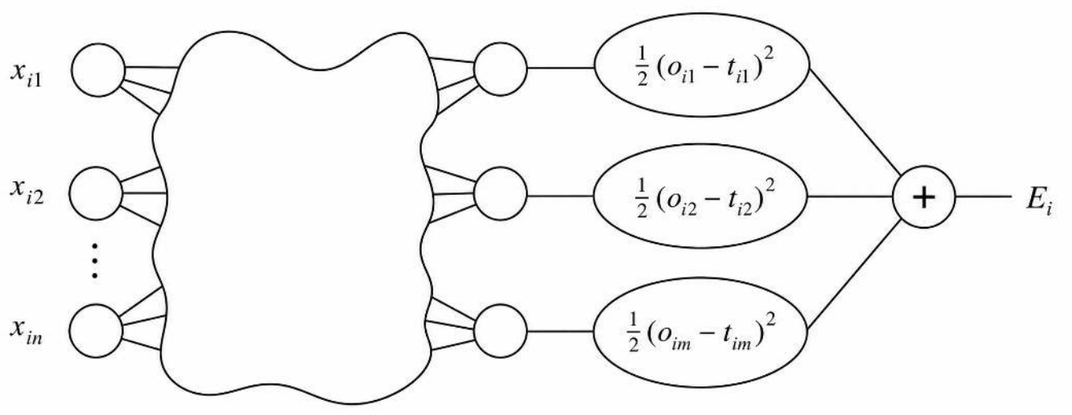
ReLU: Rectified Linear Unit.

# Training of a multilayer perceptron

- Weights are initialized with carefully selected random values
- For each training item, the predicted output is calculated (“forward pass”)



# Loss function




The network's prediction error is measured with a **loss function**.

For one training example $i$:

$$E_i = \frac{1}{2}\sum_{j=1}^{m}(o_{ij} - t_{ij})^2$$

This is the squared error loss across all **output neurons** $j$.

Across all $h$ training examples, the total loss is:

$$E = \sum_{i=1}^{h} E_i$$

Backpropagation adjusts the weights to reduce this total loss.

# Backpropagation

Backpropagation calculates the gradients of the loss function with respect to the network's weights. These gradients are then used by an optimizer, to iteratively update the weights and reduce the difference between predicted and true outputs.

```
initialize weights and biases

repeat until stopping criterion is met:

    total_error = 0

    for each training example:

        pass input forward through all layers
        compare predicted output with target value
        add error to total_error
        propagate error backward through all layers
        update weights and biases

    check whether total_error is small enough
```

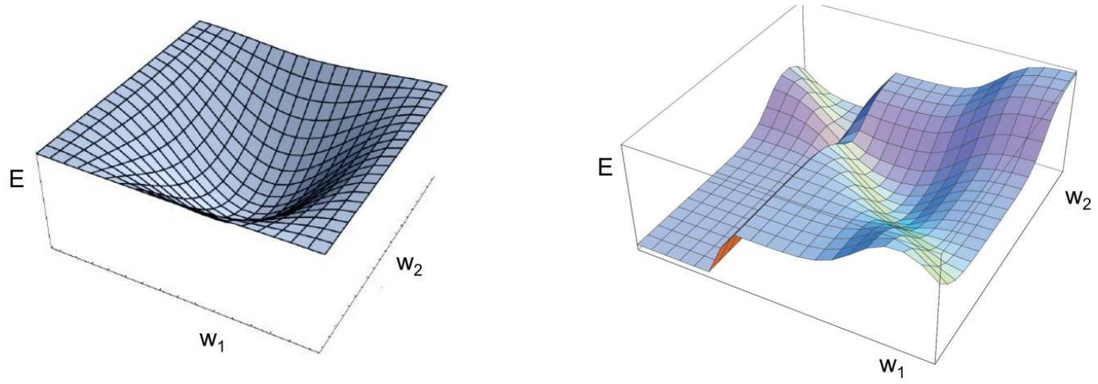
# Using the loss function to adjust weights

The loss function $E$ has to be minimized. Because it depends on the output neurons $o_j$, it automatically depends on their weights to the precedent layer(s):

$$o_j = f(s(x)_j) \text{ with } s(x)_j = \sum_k^n w_{jk} \cdot x_k$$

Thus, the weights have to be found where $E$ is minimal.

Examples of the loss function with **two weights** (simplified):



# Gradient descent

To minimize the loss function $E$ the backpropagation algorithm uses the method of gradient descent. This method searches those weights, where the vector containing the partial first derivatives of the loss function $\nabla E$ (gradient) is equal to the zero vector (minimum):

$$\nabla E = \left( \frac{\partial E}{\partial w_1}, \frac{\partial E}{\partial w_2}, \dots, \frac{\partial E}{\partial w_m} \right)$$

To adjust the weight $w_{ij}$, which connects neuron $i$ to neuron $j$, gradient descent updates the weight in the negative gradient direction:

$$\Delta w_{ij} = -a \cdot \frac{\partial E}{\partial w_{ij}}$$

$$\Delta w_{ij} = -a \cdot \sum_{j=1}^{m} (o_{ij} - t_{ij}) \cdot (-x_i) = a \cdot \sum_{j=1}^{m} (o_{ij} - t_{ij}) \cdot x_i$$

where $a$ represents the predefined learning rate. The adjusted weight is then computed as:

$$w_{ij}^{\text{new}} = w_{ij}^{\text{old}} + \Delta w_{ij}$$

> Because neural networks have many parameters, each update changes the model in many dimensions at once. The learning rate therefore has to be chosen carefully: large steps can destabilize training, while very small steps can make learning inefficient.

# Specialized NNs

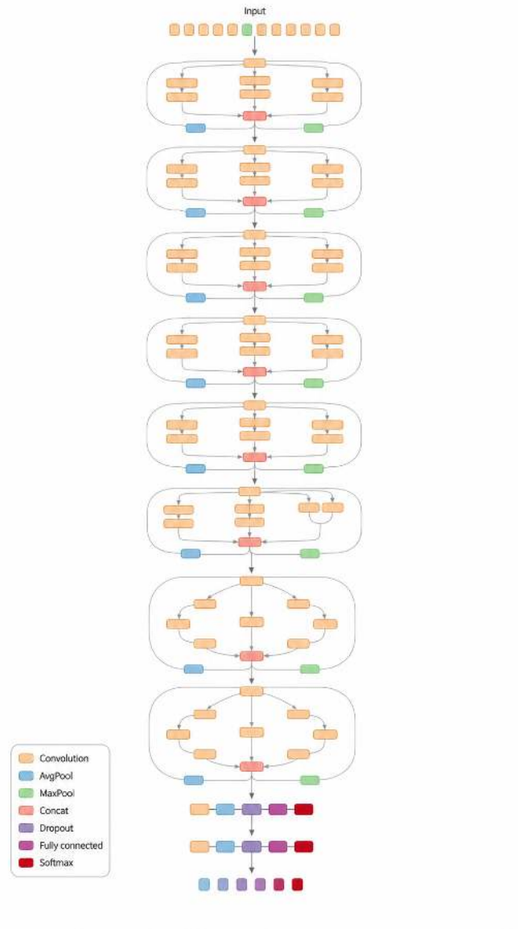
# Overview

Basic neural networks are often introduced as **fully connected feed-forward networks**. Specialized architectures go beyond this and differ in the following design choices:

| Design choice | How architectures differ |
|---|---|
| **Input structure assumed** | Different architectures are built for different data structures, such as tabular data (MLPs), spatial data and images (CNNs), sequence data (RNNs or Transformers), relational data (graph neural networks), or ; generative tasks (GANs). |
| **Neuron / unit type** | The basic unit may be a dense neuron, recurrent cell, gated LSTM/GRU cell, attention head, or a generator/discriminator module. |
| **Layer operation** | Layers may perform matrix multiplication, convolution with moving filters, recurrent state updates, self-attention, or adversarial generator/discriminator training. |
| **Connectivity pattern** | Information may flow through all-to-all dense connections, local sliding windows, temporal loops with hidden states, or token-to-token attention. |

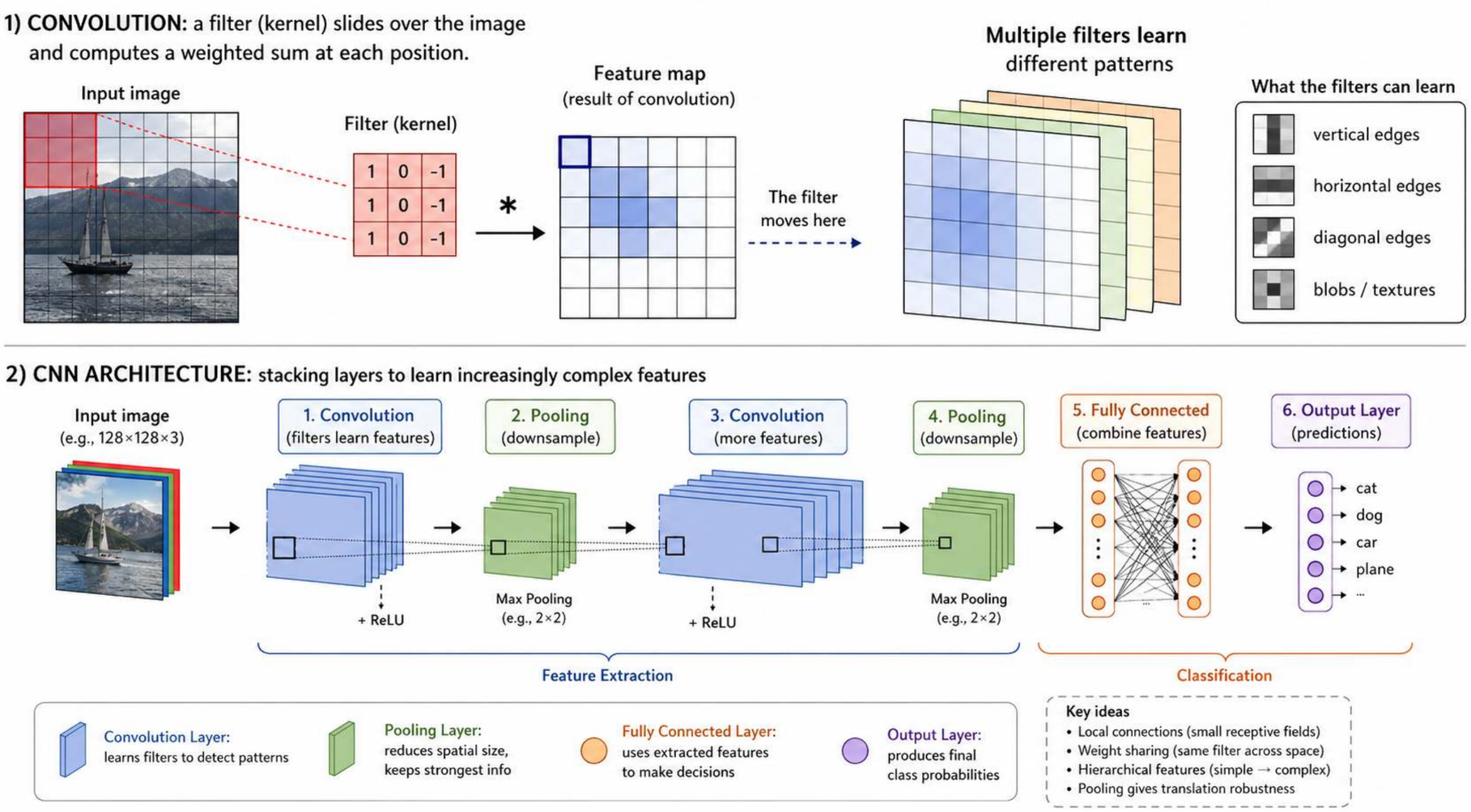
**Focus in this section**: We use **CNNs**, **RNNs**, and **Transformers** as key examples because they show three central architectural ideas:

- **Convolution through moving filters**
- **Memory through recurrence**, and
- **Attention**.




Example architecture: Inception-ResNet-v2

# Convolutional neural networks (CNN)



# Recurrent neural networks (RNN)

### CORE PRINCIPLE

An RNN processes a sequence step by step.

At each time step $t$:

- The current input $x_t$ is combined with the previous hidden state $h_{t-1}$.
- The hidden state $h_t$ is updated.
- The hidden state carries information (**memory**) forward in time.

### RECURRENCE (CORE IDEA)

$$h_t = f\left(Wx_t + Uh_{t-1} + b\right)$$

| | |
|---|---|
| $x_t$ | = input at time step $t$ |
| $h_{t-1}$ | = previous hidden state (**memory**) |
| $h_t$ | = updated hidden state |
| **W, U** | = learned weight matrices |
| $b$ | = bias term |
| $f$ | = activation function (tanh, ReLU, ...) |

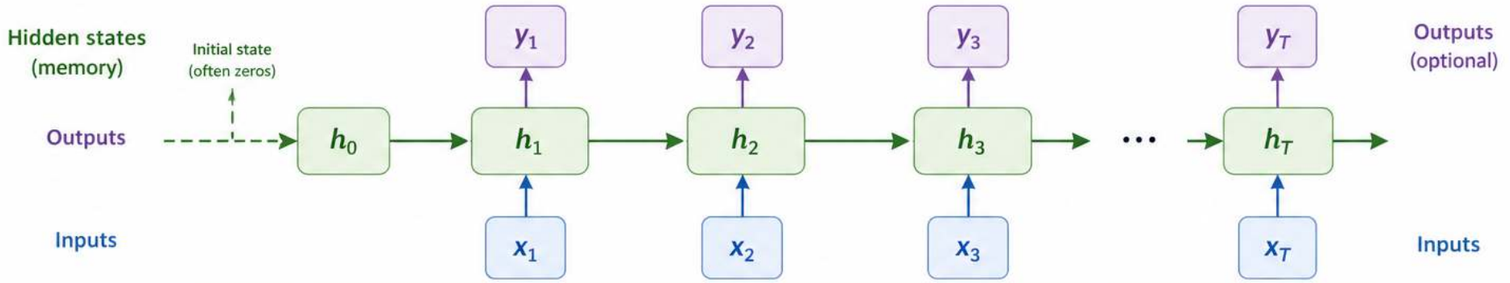
### WHAT THE HIDDEN STATE DOES

**Stores** information about what has been seen so far.

Provides **context** for understanding the current input.

Helps **solve** tasks that depend on sequential order.

### RNN ARCHITECTURE (UNROLLED OVER TIME)



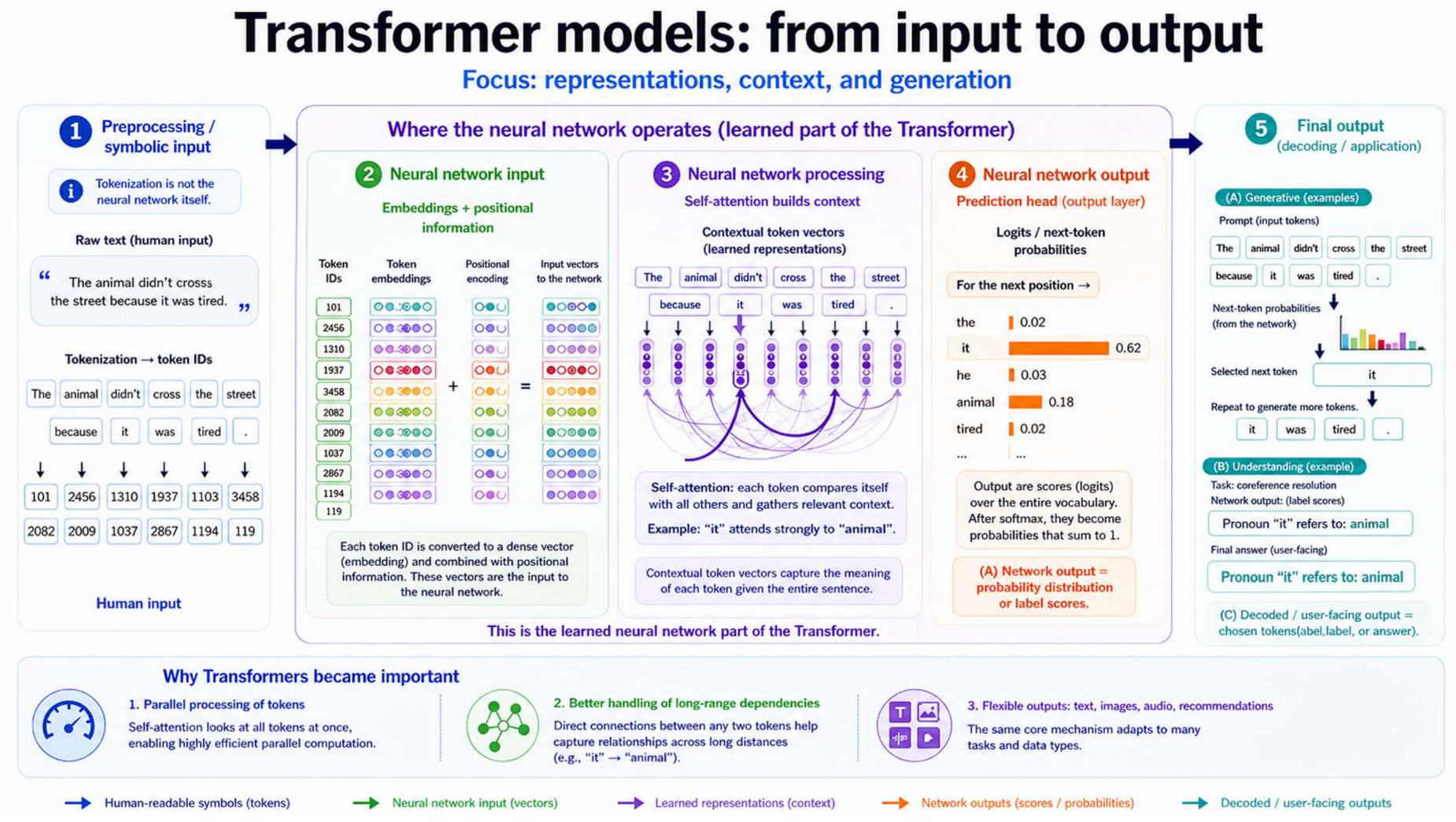

### HOW IT WORKS (STEP BY STEP)

1. At each time step $t$, input $x_t$ enters the network.
2. $x_t$ and the previous hidden state $h_{t-1}$ are combined.
3. The hidden state is updated: $h_t = f\left(Wx_t + Uh_{t-1} + b\right)$.
4. $h_t$ is passed to the next time step (memory).
5. An output $y_t$ can be produced (at each step or at the end).

### WHERE RNNs ARE USED

- Language modeling, text generation
- Speech recognition
- Time-series forecasting
- Machine translation
- ...

# Transformers



# Design and training of neural networks

# Modeling: choose the architecture and training setup

💡 **Best-practice logic** Start with choices that are mostly determined by the **data structure** and the **prediction task**.
Then tune the parts that require experimentation.

### Architecture family

| Data structure | Typical model | Inductive bias |
|---|---|---|
| 📊 Tabular data | MLP | fully connected |
| 📷 Images | CNN | local / spatial |
| 🎧 Audio | CNN / RNN / Transformer | local, sequential, attention |
| ⏱️ Time series | RNN / Transformer | temporal order |
| 📝 Text / language | Transformer | global attention |

### Output layer and loss function

| Prediction task | Output layer | Output activation | Loss function |
|---|---|---|---|
| Regression | 1 neuron | Linear | MSE |
| Binary classification | 1 neuron | Sigmoid | Binary cross-entropy |
| Multi-class classification | One neuron per class | Softmax | Categorical cross-entropy |

**Rule of thumb:** data structure determines the architecture; prediction task determines the output layer and loss.

# Training setup choices

Besides the selection of the model type, output layer, and loss function (see previous slide), these choices are usually standardized:

- **Adam** is a strong default optimizer
- Hidden layers usually start with **ReLU**

These choices require **more experimentation** to tune complexity and parameters:

1. Network structure
   *How much can the model represent?*

**Depth**
number of layers

- shallow → simple patterns
- deep → complex abstractions

**Width**
neurons per layer

Typical starts:
32, 64, 128, 256

2. Learning rate $\eta$
   *How large are the update steps?*

$$w := w - \eta \nabla_w J$$

- too high → unstable
- too low → slow

Typical range:
$10^{-3}$ to $10^{-4}$

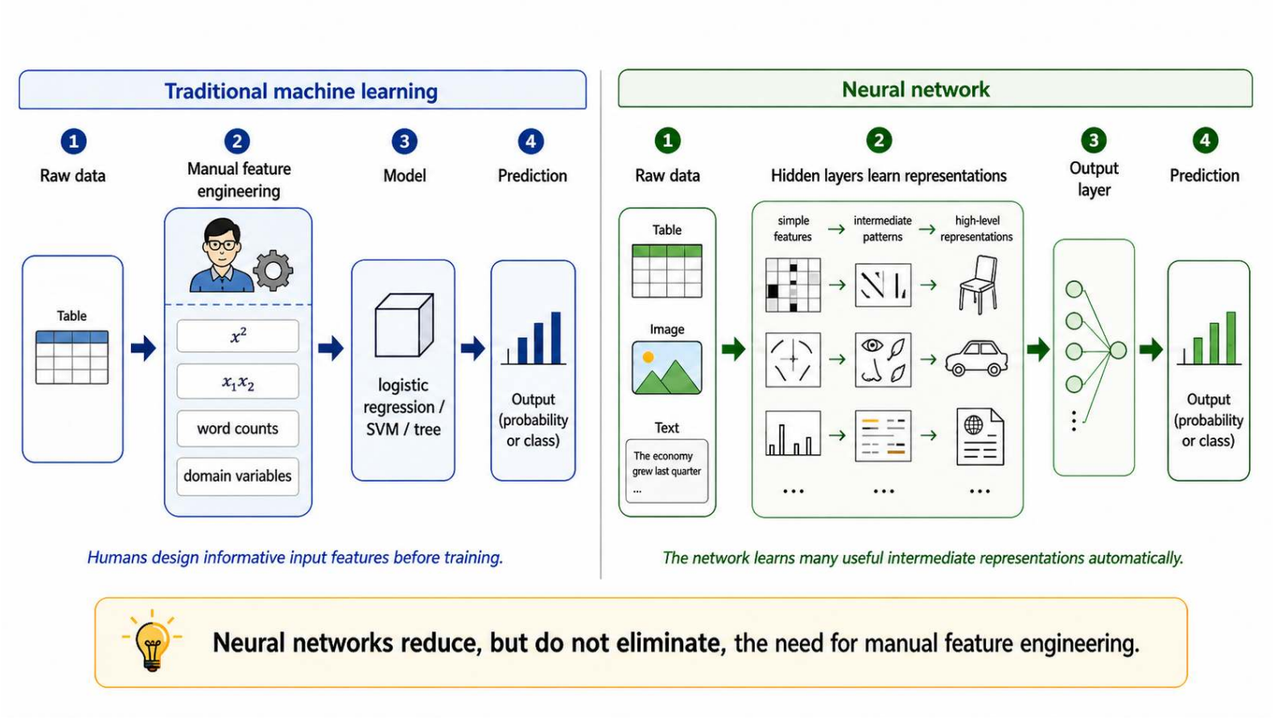
3. Regularization
   *How do we reduce overfitting?*

- **Dropout**
  randomly disables units
- **Weight decay**
  penalizes large weights
- **Early stopping**
  stops before overfitting

# Note: Neural networks and feature engineering

Traditional machine learning often relies strongly on **manual feature engineering**: humans design useful input variables before training the model. Neural networks can reduce this need because hidden layers learn **intermediate representations** from data. However, feature engineering does not disappear. We still need to decide:

- which data are provided as input
- how inputs are encoded and normalized
- whether domain-specific variables or transformations are added



# Python code example

The `sklearn` library provides basic feedforward networks, including `MLPClassifier` and `MLPRegressor`. It does not offer CNNs, RNNs, or other advanced neural networks.

```python
from sklearn.neural_network import MLPClassifier
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y)

model = MLPClassifier(
    hidden_layer_sizes=(32, 16),  # architecture (depth + width)
    activation="relu",            # hidden activation
    solver="adam",                # optimizer
    learning_rate_init=0.001,     # learning rate
    alpha=0.0001,                 # L2 regularization (weight decay)
    max_iter=200
)
model.fit(X_train, y_train)
```

# PyTorch (I)

For advanced neural networks and production environments, `pytorch` and `tensorflow` are suitable libraries.

```python
import torch

X = torch.randn(100, 10)
y = torch.sum(X, dim=1, keepdim=True)  # simple target

# TODO: specify MLP

model = MLP(
    input_dim=10,
    hidden_layers=[32, 16],   # try: [8], [64, 64], [128, 64, 32]
    output_dim=1              # regression
)

# Training setup
criterion = nn.MSELoss()   # regression
learning_rate = 1e-3       # try: 1e-1, 1e-4
optimizer = torch.optim.Adam(model.parameters(), lr=learning_rate)

# Training loop
for epoch in range(100):

    predictions = model(X)  # Forward pass
    loss = criterion(predictions, y)
```

# PyTorch (II)

```python
import torch.nn as nn

# Specify MLP
class MLP(nn.Module):
    def __init__(self, input_dim, hidden_layers, output_dim):
        super().__init__()

        layers = []
        prev_dim = input_dim

        # Depth and width
        for h in hidden_layers:
            layers.append(nn.Linear(prev_dim, h))
            layers.append(nn.ReLU())
            prev_dim = h

        layers.append(nn.Linear(prev_dim, output_dim))

        self.model = nn.Sequential(*layers)

    def forward(self, x):
        return self.model(x)

#
```

See the torch Module for inherited methods.

# Summary

- Neural networks transform inputs into outputs through **weighted connections**, **biases**, and **activation functions**.
  A single perceptron can only learn **linear decision boundaries**; multilayer networks with nonlinear activations can model more complex patterns.
- Training consists of a **forward pass**, loss calculation, **backpropagation**, and parameter updates via an optimizer such as gradient descent or Adam.
- Different neural network architectures encode different assumptions about data:
  **MLPs** for tabular data, **CNNs** for spatial data, **RNNs** for sequences, and **Transformers** for attention-based language tasks.
- Designing neural networks means choosing the architecture, output layer, loss function, learning rate, optimizer, and regularization strategy, then iteratively evaluating and adjusting these choices.

# Survey: Session 9

https://forms.gle/1vsCsqc3SzWfSX1f6

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏

# References

Zhang, A., Lipton, Z. C., Li, M., & Smola, A. J. (2023). *Dive into deep learning*. Cambridge University Press.